

Node Importance via Betweenness Centrality

Lab Computational Analytics - SoSe26

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Definition

Given a (weighted) graph $G = (V, E)$ and $s \neq t \neq v \in V$, let σ_{st} denote the number of shortest $s - t$ -paths in G and let $\sigma_{st}(v)$ denote the number of shortest $s - t$ -paths containing v . Then the centrality betweenness of a vertex $v \in V$ is defined as

$$c_B(v) := \sum_{s \neq t \in V \setminus \{v\}} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

The state-of-the-art exact algorithm by Brandes, 2001 runs in $\mathcal{O}(nm + n^2 \log n)$ time on weighted graphs and $\mathcal{O}(nm)$ time on unweighted graphs. Assuming the Strong Exponential Time Hypothesis, no truly sub-quadratic algorithm exists.

The betweenness centrality naturally induces an ordering of the nodes.

Probabilistic approach

The algorithm by Borassi and Natale, 2019 combines balanced bidirectional BFS and adaptive sampling, and estimates centrality betweenness. They provide pseudocode for the unweighted case, but the algorithm also works and should be implemented for weighted graphs.

ML approach

Dachille et al., 2026 introduce BRAVA-GNN, a Graph Neural Network approach that uses multi-hop degree mass to estimate the order of the nodes induced by betweenness centrality for unweighted graphs.

Possibly compare with ML approach by Mirakyan, 2021.

-  Borassi, Michele and Emanuele Natale (2019). “KADABRA is an ADaptive Algorithm for Betweenness via Random Approximation”. In: *ACM J. Exp. Algorithmics* 24. DOI: 10.1145/3284359.
-  Brandes, Ulrik (2001). “A faster algorithm for betweenness centrality*”. In: *The Journal of Mathematical Sociology* 25.2, pp. 163–177. DOI: 10.1080/0022250X.2001.9990249.
-  Dachille, Justin et al. (2026). *BRAVA-GNN: Betweenness Ranking Approximation Via Degree MAss Inspired Graph Neural Network*. DOI: 10.48550/arXiv.2602.09716. arXiv: 2602.09716 [cs.LG].
-  Maurya, Sunil Kumar, Xin Liu, and Tsuyoshi Murata (2019). “Fast Approximations of Betweenness Centrality with Graph Neural Networks”. In: *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*. CIKM '19. Beijing, China: Association for Computing Machinery, pp. 2149–2152. DOI: 10.1145/3357384.3358080.
-  Mirakyan, Martin (Sept. 2021). “ABCDE: Approximating Betweenness-Centrality ranking with progressive-DropEdge”. In: *PeerJ Computer Science* 7, e699. DOI: 10.7717/peerj-cs.699.